

Supporting Theory for Incremental Modality Acquisition

Frozen-LM Residual Adapters, Interference, and Layer Placement

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Abstract

This note collects only the theory needed to support the NeurIPS paper on incremental modality addition to a frozen language model. The central points are: (i) gated residual adapters preserve the base model exactly when the added modalities are inactive, (ii) composition error arises from second-order cross-terms between modality perturbations, (iii) the empirical synergy metric is a score-level proxy for this interaction, and (iv) staggered layer placement removes direct same-layer coupling and replaces it with an attenuated Jacobian-mediated interaction. We deliberately omit sidecar mixing, unrelated training objectives, and other speculative extensions.

1 Setup

Let a frozen decoder-only transformer have layers

$$h_{\ell+1} = f_{\ell}(h_{\ell}), \quad \ell = 0, \dots, L-1,$$

with text-conditioned initial state $h_0 = \text{Embed}(x_{\text{text}})$.

For each added modality m :

- a frozen encoder E_m ,
- a trainable projector P_m ,
- trainable fusion blocks $\Phi_{m,\ell}$ at selected layers $\ell \in S_m$,
- a gate $g_{m,\ell}$ controlling injection strength.

The projected modality representation is

$$z_m = P_m(E_m(x_m)).$$

At a fusion layer ℓ , the model applies pre-activation residual fusion

$$\tilde{h}_{\ell} = h_{\ell} \sum_{m \in \mathcal{M}_{\ell}(x)} g_{m,\ell}(x) \Delta_{m,\ell}(h_{\ell}, z_m), \quad h_{\ell+1} = f_{\ell}(\tilde{h}_{\ell}), \quad (1)$$

where $\Delta_{m,\ell} := \Phi_{m,\ell}(h_{\ell}, z_m)$ and $\mathcal{M}_{\ell}(x)$ is the set of active modalities on example x .

Scope. All claims below concern the frozen-backbone residual-adapter setting in Eq. (1). They do *not* claim that arbitrary fine-tuning methods preserve the base model, nor that any observed metric always admits an exact second-order explanation globally.

2 Exact Gated-Off Invariance

Definition 1 (Inactive modality). *A modality m is inactive on input x if its fusion contribution is exactly zero at every fusion layer:*

$$g_{m,\ell}(x) \Delta_{m,\ell}(h_\ell, z_m) = 0 \quad \forall \ell \in S_m.$$

This covers the cases “modality absent”, “gate forced to zero”, or any implementation that produces the zero residual when the modality is disabled.

Proposition 1 (Exact invariance of the frozen hub). *If every added modality is inactive on input x , then the model with adapters produces exactly the same hidden states and outputs as the original frozen text-only backbone on x .*

Proof. Under inactivity, the sum in Eq. (1) is identically zero, so

$$\tilde{h}_\ell = h_\ell \quad \forall \ell.$$

Therefore the recurrence reduces to

$$h_{\ell+1} = f_\ell(h_\ell),$$

which is exactly the original backbone update at every layer. By induction on ℓ , all hidden states coincide with the frozen text-only model. Since the output head is unchanged, the final logits and predictions also coincide. $\square$

Remark 1. *Proposition 1 is an exact functional statement, not a statistical one. It does not say that the model is robust to arbitrary nonzero gates; it says that when the added modality path is truly zero, the base model is preserved exactly.*

3 Second-Order Interaction of Two Modalities

To analyze composition locally, fix a hidden state $u \in \mathbb{R}^d$ at some fusion site and let $\delta_a, \delta_v \in \mathbb{R}^d$ be the net residual perturbations induced by audio and vision at that site. Let $q : \mathbb{R}^d \rightarrow \mathbb{R}$ be any smooth scalar surrogate quantity of interest, for example a negative log-likelihood, logit margin, or task score proxy.

Proposition 2 (Second-order decomposition). *Assume q is three times continuously differentiable near u . Then*

$$\begin{aligned} q(u + \delta_a + \delta_v) &= q(u) + \langle \nabla q(u), \delta_a + \delta_v \rangle + \frac{1}{2}(\delta_a + \delta_v)^\top H_q(u)(\delta_a + \delta_v) + \mathcal{O}((\|\delta_a\| + \|\delta_v\|)^3) \\ &= \underbrace{q(u) + \langle \nabla q(u), \delta_a \rangle + \frac{1}{2}\delta_a^\top H_q(u)\delta_a}_{\text{audio-only terms}} + \underbrace{\langle \nabla q(u), \delta_v \rangle + \frac{1}{2}\delta_v^\top H_q(u)\delta_v}_{\text{vision-only terms}} \\ &\quad + \underbrace{\delta_a^\top H_q(u)\delta_v}_{\text{cross-term}} + \mathcal{O}((\|\delta_a\| + \|\delta_v\|)^3). \end{aligned} \tag{2}$$

Proof. Apply the multivariate second-order Taylor expansion of q at u with increment $\delta_a + \delta_v$:

$$q(u + \eta) = q(u) + \langle \nabla q(u), \eta \rangle + \frac{1}{2}\eta^\top H_q(u)\eta + \mathcal{O}(\|\eta\|^3), \quad \eta := \delta_a + \delta_v.$$

Expanding the quadratic term gives

$$\frac{1}{2}(\delta_a + \delta_v)^\top H_q(u)(\delta_a + \delta_v) = \frac{1}{2}\delta_a^\top H_q(u)\delta_a + \frac{1}{2}\delta_v^\top H_q(u)\delta_v + \delta_a^\top H_q(u)\delta_v,$$

since $H_q(u)$ is symmetric. Substituting this identity yields Eq. (2). $\square$

Remark 2 (Interpretation). *The cross-term $\delta_a^\top H_q(u) \delta_v$ is the leading source of local non-additivity. If it is large and negative for a score-like q , joint composition is worse than additive expectation; if it is positive, the two modalities reinforce each other.*

3.1 Gate scaling

Write each perturbation as $\delta_m = g_m \hat{\delta}_m$, where $\hat{\delta}_m$ is the direction produced by the modality path and $g_m \geq 0$ is the gate.

Corollary 1 (Quadratic scaling of cross-modal interference). *Under the notation above, the leading interaction term scales as*

$$\delta_a^\top H_q(u) \delta_v = g_a g_v \hat{\delta}_a^\top H_q(u) \hat{\delta}_v.$$

Thus the cross-modal interaction is second-order in gate magnitude, while each unimodal first-order term scales only linearly in its own gate.

Proof. Substitute $\delta_a = g_a \hat{\delta}_a$ and $\delta_v = g_v \hat{\delta}_v$ into the bilinear cross-term:

$$\delta_a^\top H_q(u) \delta_v = (g_a \hat{\delta}_a)^\top H_q(u) (g_v \hat{\delta}_v) = g_a g_v \hat{\delta}_a^\top H_q(u) \hat{\delta}_v.$$

□

Remark 3. *Corollary 1 explains why small initial gates can stabilize early training: they suppress the interaction term faster than they suppress the individual linear signals.*

4 Synergy as an Empirical Proxy

In experiments we do not usually optimize or report q directly. Instead we observe four task metrics:

$$y_0 \quad (\text{text-only}), \quad y_a \quad (\text{audio-only}), \quad y_v \quad (\text{vision-only}), \quad y_{av} \quad (\text{both}).$$

Definition 2 (Synergy). *The synergy of the two-modality system is*

$$\text{syn}(y) := y_{av} - (y_a + y_v - y_0). \quad (3)$$

Definition 3 (Gain vs best single). *The operational composition gain is*

$$\Delta^{\text{best}} := y_{av} - \max(y_a, y_v). \quad (4)$$

Meaning. Equation (3) compares the observed composed score to the score predicted by exact additivity. Equation (4) answers the practical question of whether composition beats the strongest unimodal system.

Proposition 3 (Surrogate-level synergy identity). *Let q_0, q_a, q_v, q_{av} be the values of a smooth scalar surrogate under text-only, audio-only, vision-only, and composed conditions, all linearized at the same base hidden state u . Then*

$$q_{av} - (q_a + q_v - q_0) = \delta_a^\top H_q(u) \delta_v + \mathcal{O}((\|\delta_a\| + \|\delta_v\|)^3). \quad (5)$$

Proof. Apply Proposition 2 to $q_{av} = q(u + \delta_a + \delta_v)$ and the corresponding second-order expansions for $q_a = q(u + \delta_a)$ and $q_v = q(u + \delta_v)$. Subtracting $q_a + q_v - q_0$ cancels all constant, gradient, and unimodal quadratic terms, leaving only the mixed quadratic term plus third-order remainder. □

Remark 4. *For discrete metrics such as exact-match accuracy, Eq. (5) is not an exact theorem. The experimental synergy in Eq. (3) should be read as an empirical proxy for the same interference phenomenon, not as the Hessian cross-term itself.*

5 Why Same-Layer Fusion is More Fragile

5.1 Same-layer coupling

Suppose both modalities inject at the same layer ℓ :

$$\tilde{h}_\ell = h_\ell + \delta_{a,\ell} + \delta_{v,\ell}.$$

Then the next frozen block f_ℓ receives the *sum* of both perturbations simultaneously. By Proposition 2, the local interaction at this layer is governed by the direct mixed term

$$\delta_{a,\ell}^\top H_{f_\ell}(h_\ell) \delta_{v,\ell},$$

where $H_{f_\ell}(h_\ell)$ denotes the appropriate second derivative of the frozen layer map or of a scalar surrogate composed with that map.

5.2 Disjoint-layer fusion

Now suppose audio injects at layer ℓ_a and vision injects later at $\ell_v > \ell_a$, with no shared injection at intermediate layers. Then

$$\tilde{h}_{\ell_a} = h_{\ell_a} + \delta_{a,\ell_a}, \quad h_{\ell_v} = F_{\ell_a \rightarrow \ell_v}(h_{\ell_a} + \delta_{a,\ell_a}),$$

where

$$F_{\ell_a \rightarrow \ell_v} := f_{\ell_v-1} \circ \dots \circ f_{\ell_a}.$$

Linearizing at h_{ℓ_a} gives

$$h_{\ell_v} \approx F_{\ell_a \rightarrow \ell_v}(h_{\ell_a}) + J_{\ell_a \rightarrow \ell_v} \delta_{a,\ell_a},$$

where $J_{\ell_a \rightarrow \ell_v}$ is the Jacobian of the frozen backbone between the two injection sites. Vision then injects

$$\tilde{h}_{\ell_v} = h_{\ell_v} + \delta_{v,\ell_v}.$$

Proposition 4 (Direct cross-term elimination under disjoint layers). *If two modalities use disjoint layer sets, then there is no direct same-layer mixed quadratic term of the form $\delta_{a,\ell}^\top H_{f_\ell}(h_\ell) \delta_{v,\ell}$ at any fusion layer. The leading interaction is instead Jacobian-mediated:*

$$\delta_{v,\ell_v}^\top H_{f_{\ell_v}}(h_{\ell_v}) (J_{\ell_a \rightarrow \ell_v} \delta_{a,\ell_a}), \quad (6)$$

or the symmetric analogue when vision precedes audio.

Proof. At any audio-only layer $\ell \in S_a \setminus S_v$, the perturbation entering the frozen block is $h_\ell + \delta_{a,\ell}$, so no term involving $\delta_{v,\ell}$ exists. Similarly, at any vision-only layer $\ell \in S_v \setminus S_a$, the perturbation entering the frozen block is $h_\ell + \delta_{v,\ell}$, so no direct same-layer mixed term exists there either. The first point at which audio can influence the vision interaction is through the transformed hidden state that arrives at a later vision layer, which is given to first order by $J_{\ell_a \rightarrow \ell_v} \delta_{a,\ell_a}$. Substituting this transformed perturbation into the mixed term at the later vision layer yields Eq. (6). $\square$

Corollary 2 (Interaction count). *With M modalities sharing the same layer, the number of direct mixed pairs at that layer is $\binom{M}{2}$. With disjoint layer sets, the number of direct same-layer mixed pairs is zero.*

Proof. In the shared-layer case, every unordered pair of active modality perturbations contributes one mixed quadratic term. The number of unordered pairs among M modalities is $\binom{M}{2}$. Under disjoint layer sets, no layer contains more than one modality perturbation, so no such same-layer pair exists. $\square$

Remark 5. *Disjoint placement does not eliminate all interaction. It only removes the most direct one. The remaining interaction is mediated by the frozen backbone Jacobian, which is typically weaker and easier to manage with gates and training.*

6 Diagnostics Used in Practice

For a layer-level probe at layer ℓ , define:

$$y_{0,\ell}, \quad y_{a,\ell}, \quad y_{v,\ell}, \quad y_{av,\ell}.$$

Definition 4 (Layer additivity error). *The layer additivity error is*

$$\epsilon_\ell := \frac{|y_{0,\ell} - y_{a,\ell} - y_{v,\ell} + y_{av,\ell}|}{|y_{a,\ell} - y_{0,\ell}| + |y_{v,\ell} - y_{0,\ell}| + \varepsilon}, \quad (7)$$

with $\varepsilon > 0$ for numerical stability.

Interpretation. Small ϵ_ℓ means the layer behaves close to additive composition; large ϵ_ℓ indicates a high degree of non-additive interaction.

Why use both ϵ_ℓ and $\Delta_\ell^{\text{best}}$. The normalized error in Eq. (7) measures stability of composition. The gain in Eq. (4) measures usefulness of composition. A strong fusion layer should ideally have both low ϵ_ℓ and high $\Delta_\ell^{\text{best}}$.

7 What These Results Support in the Paper

The results above support four concrete claims:

1. **Functional invariance.** Added modalities do not alter the frozen text hub when their residual paths are inactive (Proposition 1).
2. **Interference has a specific local source.** The leading error of joint composition is the mixed second-order term (Proposition 2).
3. **Synergy is the right empirical summary.** The experimental synergy metric is the observable proxy for this mixed interaction (Proposition 3).
4. **Layer placement matters structurally.** Shared layers create direct coupling; disjoint layers remove it and replace it with a weaker transformed interaction (Proposition 4).

What these results do not claim. They do not prove that joint performance must exceed the best unimodal result, that staggered layers always outperform shared layers on every backbone, or that a global discrete metric is exactly equal to a local Hessian quantity. Those are empirical questions.